

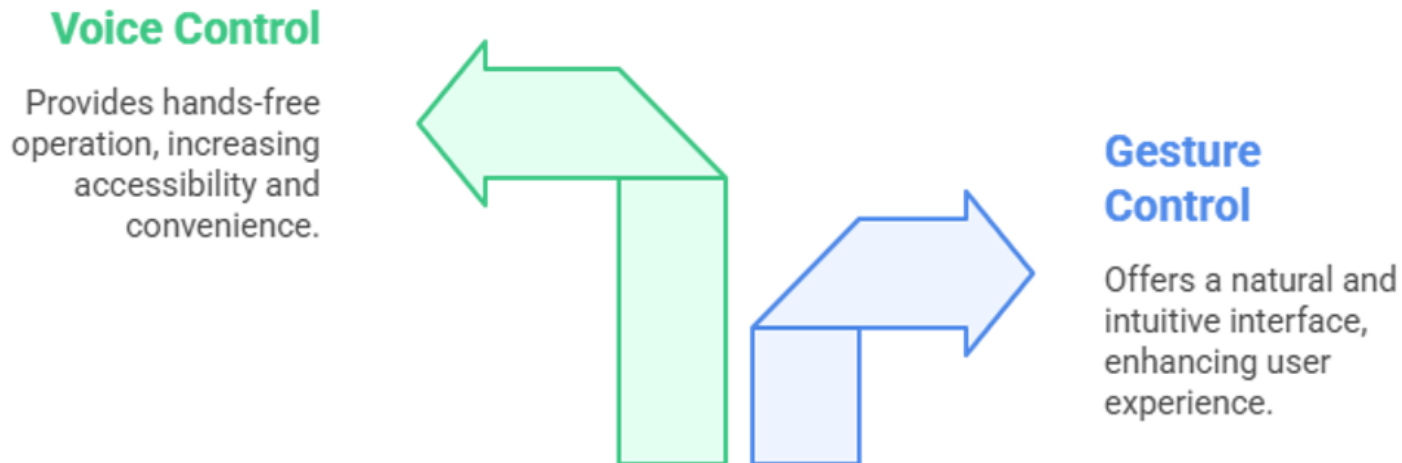
Artificial Intelligence Project

Gesture + Voice Controlled Robot Movement

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Problem Statement

- Manual control of robots using remote or wired systems lacks intuitiveness and limits accessibility. There is a need for a more natural and intelligent way to control robotic movement using human-friendly interfaces like gestures and voice.

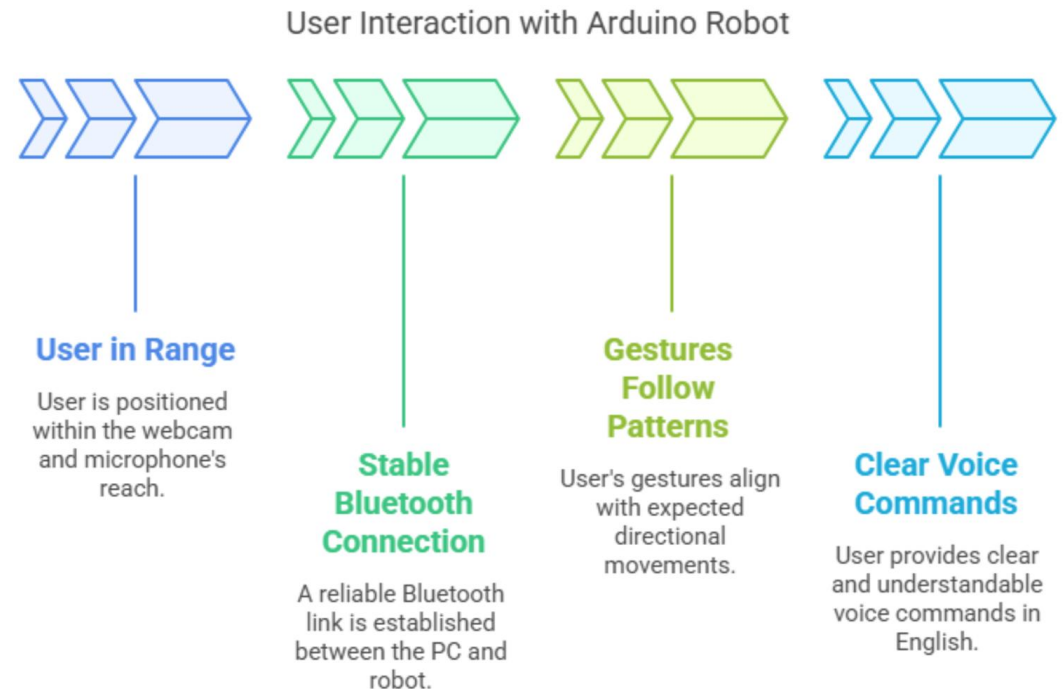


Objectives

- Develop a hybrid robot that can be controlled by both hand gestures and voice commands.
- Prioritize gesture-based control while allowing voice fallback when gesture input is neutral.
- Implement real-time hand tracking using MediaPipe and voice command interpretation using an LLM.

Assumption

- The user is in range of the webcam and microphone.
- A stable Bluetooth connection is established between the PC and the Arduino robot.
- The gestures follow expected directional patterns (up, down, left, right).
- Voice commands are given clearly and in English.



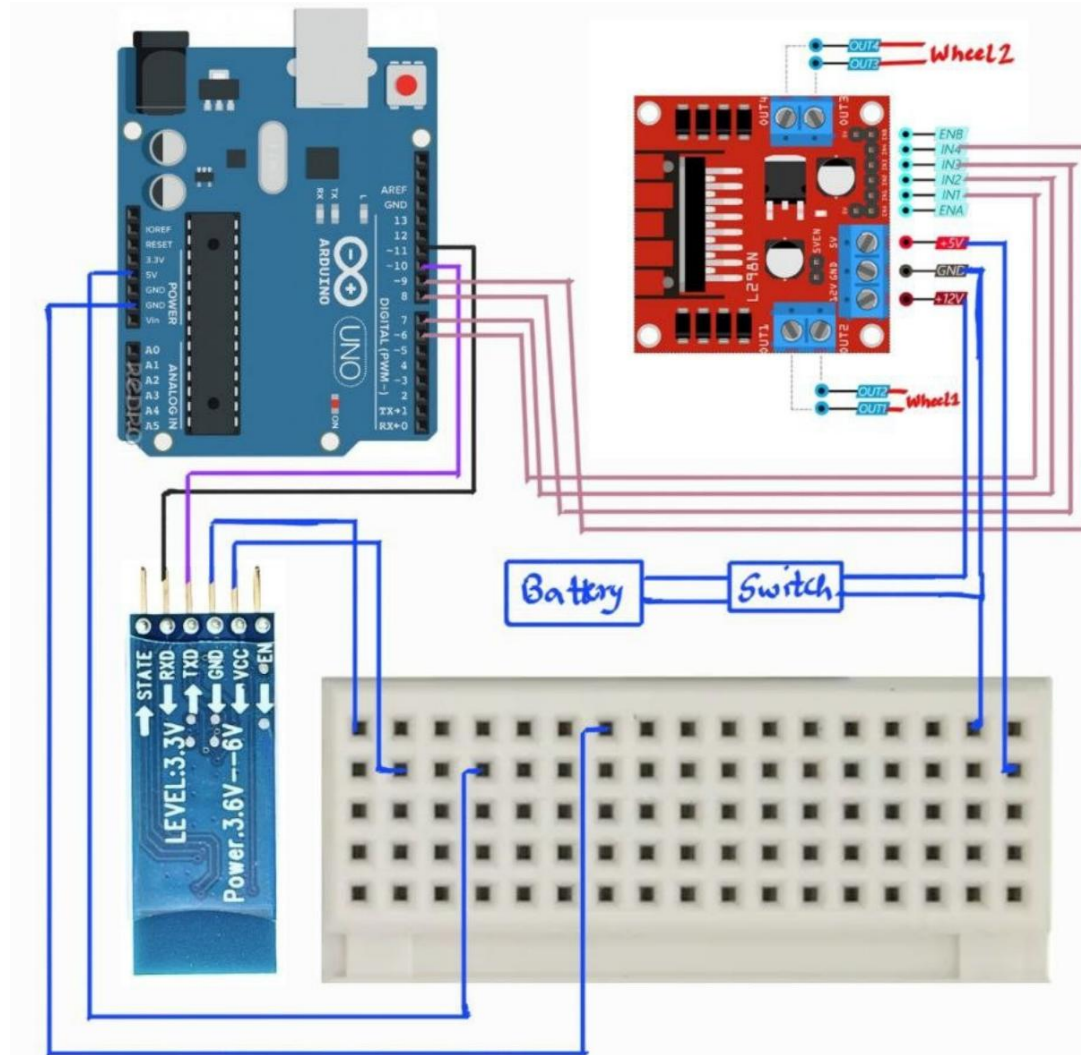
Agent Introduction

- The robot system is designed as an intelligent agent that receives perceptual input through a webcam and microphone.
- It acts based on gesture or voice interpretation to perform movement commands using motors via an Arduino controller, sends signals via bluetooth module (HC-05).
- Gesture input is prioritized over voice unless the gesture indicates 'neutral'.

AI Module Used

- MediaPipe (Google): For real-time hand gesture recognition.
- OpenCV: For processing video frames and detecting gestures visually.
- SpeechRecognition + PyAudio: For capturing and converting voice input to text.
- GitHub-hosted OpenAI GPT 4.1 Model: For interpreting natural language voice commands into actionable robot movement instructions.

Circuit Diagram



Our Learning

- Learned integration of multiple AI tools (vision, NLP, voice) into a single real-time application.
- Understood the significance of prioritizing control logic based on user behavior.
- Gained hands-on experience with Arduino-Bluetooth interfacing and AI-driven control systems.

References

- [MediaPipe Documentation by Google](#)
- [OpenCV Python Tutorials](#)
- [Speech Recognition Library](#)
- [OpenAI / GitHub-hosted GPT 4.1 model endpoint](#)
- [Arduino Motor Control Docs](#)